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Information Extraction from Scanned Invoice Documents Using Deep Learning Methods

Ufuk İlke Avcı
Dep. of Computer Engineering
Yeditepe University
İstanbul, Turkey
ufukilkeavci@gmail.com

Dionysis Goularas
Dep. of Computer Engineering
Yeditepe University
İstanbul, Turkey
goularas@cse.yeditepe.edu.tr

Emin Erkan Korkmaz
Dep. of Computer Engineering
Yeditepe University
İstanbul, Turkey
ekorkmaz@cse.yeditepe.edu.tr

Barış Deveci
Intecon Informatics and Concultancy
İstanbul, Turkey
baris.deveci@intecon.com.tr

***Abstract*—In this paper, we explore innovative approaches in the field of information extraction from scanned invoice documents using deep learning methods. Our study makes significant contributions in three key areas. Firstly, we introduce a novel organizational method for labeling invoices, designed to enhance the efficiency and accuracy of data extraction. This lays a foundation for future research in this domain. Secondly, we break new ground by classifying a larger number of classes, 57 in total, far exceeding the typical 8-10 classes usually addressed in existing literature. This comprehensive classification enables a more detailed and nuanced understanding of invoice data. Lastly, we present our experimentation with various deep learning architectures, including Graph Convolutional Network (GCN), LayoutLMv1, and LayoutLMv3. Notably, our findings reveal promising, albeit preliminary results for Graph Convolutional Networks (GCN), an architecture that is not pre-trained, suggesting potential for further exploration and development in this area.**

***Index Terms*—Transformers, Graph Convolutional Networks, Key Information Extraction, LayoutLM**

I. Introduction

Extracting information from invoices is an important task in finance and administration as it facilitates organizations in processing, organizing, and analyzing financial data efficiently [1]. In recent years, the advancement of deep learning techniques, particularly in Natural Language Processing (NLP) and computer vision, has greatly enhanced information extraction systems [2]. This paper presents a study on training and testing Graph Convolutional Networks (GCN), LayoutLMv1 and LayoutLMv3 models in the domain of information extraction from invoices. The study focuses on sequential node classification, sequence classification and sequence classification tasks respectively.

Invoice documents are semi-structured documents with textual and non-textual information spatially located on the page [1]. Traditional approaches to information extraction, such as rule-based methods and shallow machine learning techniques, are often unable to handle the complexity of these texts [3]. As a result of their ability to model complex relationships between features and create suitable data representations, deep learning methods have emerged as a potential alternative [4].

In this study, we propose an approach to extract information from scanned invoice documents by utilizing the capabilities of LayoutLMv1 [5] and LayoutLMv3 [6] models. These models combine natural language processing (NLP) and computer vision techniques to capture both textual and layout information. Also GCNs [7] are utilized for node classification tasks. This enables us to capture the relationships between various elements in the document, thereby improving the overall extraction performance.

The evaluation of models is conducted using a set of around 1450 invoice documents taken from Intecon company. Moreover, the success of the approach and potential areas for future research and development are analyzed.

In this study, we also present a method for organizing and labeling invoices by attempting to detect all elements in an invoice. This novel approach is designed to be adaptable and highly efficient, significantly enhancing the accuracy and effectiveness of data extraction in invoice processing. Tailored to address the diverse formats and data types found in invoices, this method not only serves as a robust tool for current research but also lays a foundational framework for future advancements in automated invoice processing using deep learning techniques.

This paper is organized as follows: In Section 2 we present the related works. In Section 3, we present the methodology used in this work. Section 4 shows the deep learning models employed in our research. In Section 5 we describe the implementation. Finally, in Section 6, we summarize our key findings and discuss potential avenues for future research in this domain and in Section 7 we conclude our research.

II. Background

This study focuses on innovative deep learning methods for extracting information from scanned invoice documents.

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LayoutLM is a pretrained model that integrates text and layout information to achieve state-of-the-art results in tasks such as form comprehension and document classification [5]. EATEN (Entity-aware Attention Text Extraction Network) is an end-to-end system that enhances robustness and performance compared to traditional OCR methods by directly extracting Text of Interest (ToI) from images [8]. The Attend, Copy, Parse (ACP) architecture utilizes end-to-end training without the need for word-level labels, surpassing traditional systems [9]. Robust Layout-aware Information Extraction combines BERT and graph convolutional networks to analyze text and font properties for document information extraction [10]. PICK utilizes transformers and graph convolutional networks to analyze sequences of words, cropped images, and spatial relations, achieving high accuracy [11]. Table Detection in Invoice Documents by Graph Neural Networks classifies entities in invoices as tables using a custom GCN model, achieving a 0.70 F1-score [12]. Other notable studies include CloudScan, an LSTM-based invoice analysis system that generalizes to various invoice layouts without initial configuration, achieving an F1-score of 0.84 [13]. A fully convolutional network accurately segments document images into figures, tables, and paragraphs [14]. The part-based modeling approach for invoice parsing utilizes a two-stage structure for invoice section detection and integration, which has been proven effective in automated invoice processing [15]. Deep Learning for Automatic Sale Receipt Understanding proposes a dual-check process using Deep Convolutional Neural Networks to extract high-confidence information from sales receipts [16]. Finally, OCR technology remains essential for character identification in scanned documents, with various segmentation and classification methods demonstrating their effectiveness in information extraction [17].

III. Analysis and Design

A. Dataset

The dataset is used to train the models and consists of 1200 invoice document pages as PNG images. These images were labeled by selecting the locations of the desired fields in invoices. After labeling images, a few preprocessing steps are applied to the labeled data before feeding and training models with this dataset. The first step of preprocessing is to extract textual information from document images using Tesseract OCR. TesseractOCR provides information on words in a document, including their spatial locations represented by rectangular bounding boxes. After all data gathered from OCR, words are processed to split them into word pieces, and then word context vectors are calculated. After preprocessing, the data is obtained as a sequence of pairs of word context vectors and spatial locations, with the class name for each corresponding word utilized as the output data.

For the classification of words, 57 different classes are utilized. These classes are categorized into three main groups: Main classes, auxiliary classes, and non-used classes. Main classes are defined as the information represented in an invoice that is required to be extracted. Auxiliary classes are the titles or descriptions in an invoice and serve to detect the main classes. Non-used classes refer to information found in invoices that is not essential for further analysis. When evaluating the model results, 57 classes, including an additional class named OTHER, were grouped to analyze the results based on the regions of the invoice. In Figure 1, the regions of an invoice are shown. Classes are grouped for evaluation based on the region specified in the invoice. These groups are defined as table group, header group, summary group, and VAT table group.

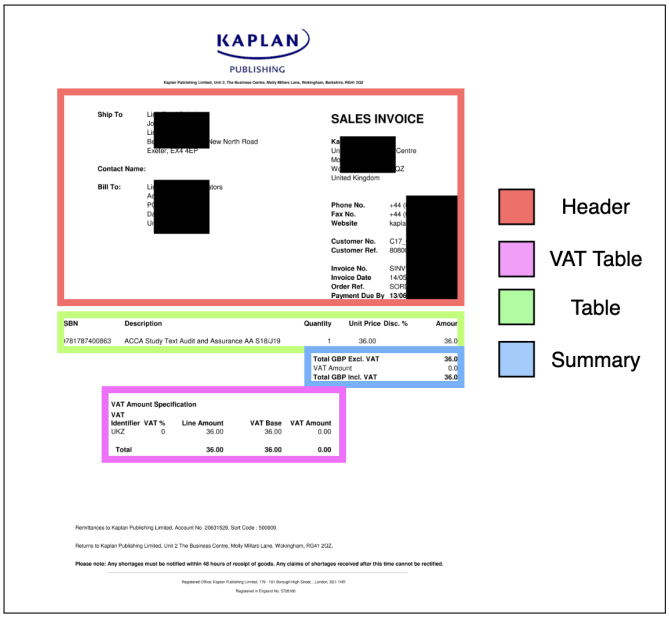


Fig. 1. Representation of class regions in an invoice

B. Definition of Classes

The classes are categorized in five groups. These classes are shown in table I

The first group is the header group, which includes classes such as invoice customer information, invoice date, invoice due date, invoice number, invoice purchase order (PO) number, supplier name, supplier address, supplier fax number, supplier phone number, and supplier Value Added Tax (VAT) number. As can be seen from the included classes, this group is related to general information about invoices and their parties.

The second selected group is the table group. Instances of this group contain information about products or services. Classes belonging to this group can be listed as follows: invoice line item number, invoice line number, invoice line quantity, invoice line subtotal, invoice line unit price, invoice line VAT, and invoice line VAT rate.

The third group is the VAT table group. This group contains the fewest class instances compared to the others. Information stored in instances of these classes describes details about taxes. VAT table header, VAT table code, VAT table rate, VAT table subtotal, and VAT table VAT classes are considered in this group.

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TABLE I
CLASSIFICATION OF CLASSES IN INVOICE DOCUMENTS

Class Group	Class Names
Header Group	invoice customer info, invoice date, invoice PO number, supplier phone number, invoice due date, invoice number, supplier fax number, supplier name, supplier vat number, supplier address
Table Group	invoice line vat rate, invoice line unit price, invoice line item no, invoice line quantity, invoice line subtotal, invoice line number, invoice line vat
VAT Table Group	invoice vat table code, invoice vat table header, invoice vat table vat, invoice vat table rate, invoice vat table subtotal
Summary Group	invoice summary vat rate, invoice summary vat, invoice summary total, invoice summary subtotal
Other Classes	invoice date label, invoice footer, invoice information header, etc.

The fourth group is the summary group and contains invoice summary information in its instances. Classes in this group include invoice summary subtotal, invoice summary total, invoice summary VAT, and invoice summary VAT rate.

Finally, the fifth group contains 32 other classes that are labeled in the invoices. These classes are created to retrieve more information from the model and deliver it to other downstream tasks.

C. Tokenization and Vectorization

After document images are scanned using OCR, raw text data is obtained. In order to tokenize and vectorize the text, WordPiece method is used [18] . This method first takes the entire text as input and then splits it into word pieces that were previously defined by analyzing a large text corpus. As a continuation of the tokenization process, these word pieces are converted into vectors of length 768.

IV. MODELS

A. Custom Graph Convolutional Network

To utilize Graph Convolutional Network (GCN) architecture for information extraction from invoice documents, a GCN model was developed. The model was designed as a union of two graph convolutional layers serving as the feature extraction module and two fully connected layers serving as the classifier module of the model. To convert document data into a graph form, the nodes of the graph are constructed as context vectors of words and each node of every word is connected to the nearest words.

The model is designed to predict which class the words belong to. The output from the model is obtained as a graph that has the same node connections as the input graph. Instead of word vectors, the nodes store class information. The model architecture is illustrated in Figure 2.

As mentioned earlier, invoice images must be preprocessed to create a graph representation of the invoice page for inputting data into the GCN model. To represent an image as a graph, an algorithm was developed to create graph nodes

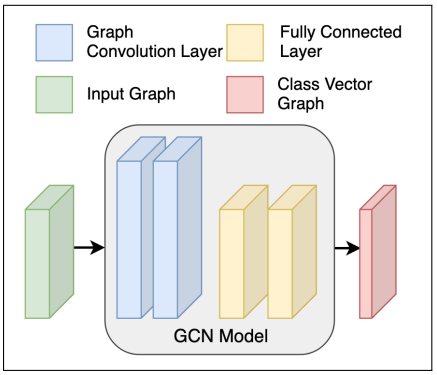


Fig. 2. Architecture of custom GCN model.

from words and edges based on spatial locations in the image. Connections between nodes are determined based on the following conditions. If two words are aligned horizontally or vertically and the words are neighbors, an edge is constructed between the nodes of these two words. In Algorithm 1, adjacency matrix for graph conversion is explained. Here, "W" represents a list of words, and "P" represents a list of positions in the form [x0, y0, x1, y1]. The algorithm constructs an adjacency matrix of a graph for initializing a graph object. Word embeddings are calculated using the tokenizer module of the BERT model by converting a list of words into a single string with spaces for separation. Horizontal-Alignment (HA) and Vertical-Alignment (VA) procedures check if the bounding boxes of two words overlap in a single axis by more than half of the size of the shorter bounding box edge, and if there is no other word between the two words.

Algorithm 1 Graph construction algorithm

$W_i \leftarrow$ word i in string format
$MBR_i \leftarrow$ Minimum bounding rectangle of word i
$M \leftarrow$ number of words
$A \leftarrow N \times N$Adjacency matrix of the graph
for $i = 1$ to M **do**
  for $j = 1$ to M **do**
    if $i \neq j$ **then** // not the same word
      if MBR_i is neighbour with MBR_j **then**
        $A[i, j] \leftarrow 1$
      end if
    end if
  end for
end for

B. LayoutLM Models

LayoutLMv1, a transformer-based model, is employed for its ability to integrate NLP and computer vision techniques. This model enhances text extraction by incorporating layout embeddings which capture both textual content and spatial arrangement in documents. In our study, LayoutLMv1 was adapted to classify words into 228 classes, focusing on the specific requirements of invoice data extraction [5].

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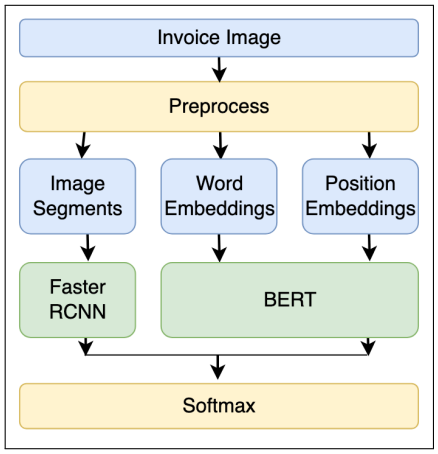


Fig. 3. LayoutLM architecture.

LayoutLM is a pretrained language model that can be used to recognize and understand the layout of document images, including PDFs, scans, and pictures [5]. The model is built on top of the well-known BERT architecture [19], but it has been expanded to include layout embedding, which records details on the physical positioning of text and graphics within a document. This enables the model to perform a variety of document analysis-related tasks, such as word recognition, table detection, and form recognition.

LayoutLMv3 extends the capabilities of LayoutLMv1 by incorporating visual embeddings, enabling a more nuanced understanding of document layouts. Our research utilized this model to investigate its effectiveness in the nuanced task of invoice data extraction.

In order to enhance performance on tasks such as information extraction, named entity recognition, and document categorization, LayoutLMv3 is a state-of-the-art pretrained model for document understanding tasks [6]. LayoutLMv3 offers several enhancements to achieve better performance on benchmark datasets, building on the success of its predecessors LayoutLM [5] and LayoutLMv2 [20].

In conclusion, LayoutLMv3 incorporates the Visual Layout Transformer and a multi-modal fusion mechanism, resulting in improved performance on benchmark datasets and an expanded scope of document understanding tasks. Both LayoutLMv1 and LayoutLMv3 build upon the BERT model for document understanding tasks.

V. Implementation

All operations involving models were conducted on a computer running Ubuntu 20.04 OS, equipped with a 12-core 3.7 GHz CPU, Nvidia RTX3090 GPU, and 64 GB of RAM. For creating a custom GCN model, the torch-geometric library was used in conjunction with PyTorch. For LayoutLM models, PyTorch and the Transformers library were used.

For the implementation of a custom GCN model, the GCNConv [21], GENConv [22] and GCN2Conv [23] were tested as graph convolution layers.

VI. Results and Discussion

A. Custom Graph Convolutional Network

After training the GCN model with 1200 invoices, it was tested on a dataset of 250 invoices. While some classes were correctly predicted in certain instances, the model generally had low rates of accurate predictions. No discernible pattern in the errors of the model was observed. In Table II, the average precision, recall, F1-score, and accuracy values are provided for the grouped classes.

TABLE II
GCN Grouped Metric Averages

Group	Precision	Recall	F1	Accur.	Balanced Accur.
Header	0.32	0.40	0.30	0.97	0.67
Table	0.46	0.41	0.37	0.98	0.66
Summary	0.25	0.34	0.21	0.99	0.62
VAT Table	0.45	0.39	0.38	0.97	0.69
Other	0.45	0.39	0.38	0.97	0.67
Average	**0.38**	**0.40**	**0.32**	**0.98**	**0.45**

While the GCN model achieves a high precision of 0.87 for classifying invoice PO numbers, its overall performance is low, with an F1-score below 0.5. Precision for most classes is extremely low, with the invoice summary VAT and total classes achieving 0.04 and 0.03, respectively. The model performs notably better on tabular data, with precision values of 0.93, 0.87, and 0.89 for invoice line unit price, quantity, and item description classes, respectively. Low success rates in VAT table classes are likely due to the limited sample size. Overall, the model is more effective with tabular structured data but exhibits inconsistent results across different classes.

B. LayoutLM

Similar to the GCN model, the LayoutLMv1 model used the same test dataset for evaluation after training with the same dataset. A confusion matrix was then created to assess the model's performance across different classes. The model predictions are mostly correct. Although the erroneous predictions exhibit an irregular distribution, the total error, which is significantly lower than the correct answers, positively impacts the criteria utilized for model evaluation. In Table III which is created for the class group criteria, it can be seen that the precision, recall, and F1-score values are significantly higher compared to the GCN model. While the model performance remains below its average for the header group, high success has been achieved in the table and summary groups. In addition, model evaluation cannot be based solely on accuracy. As can be seen by evaluating Table II and Table III, the accuracy value does not change consistently with the model's performance. The reason for this is that the accuracy criterion uses the true negative value in the calculation. Having many classes in the dataset leads to a significant increase in the true negative value used in model evaluation compared to other values. When the accuracy criterion is calculated with a high true negative value, the impact of true positive, false positive, and false negative values on the result decreases

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significantly. This brings the models closer to each other in terms of accuracy.

TABLE III
LAYOUTLMV1 GROUPED METRIC AVERAGES

Group	Precision	Recall	F1	Accur.	Balanced Accur.
Header	0.56	0.80	0.61	0.99	0.88
Table	0.73	0.79	0.73	1.00	0.86
Summary	0.78	0.81	0.79	1.00	0.90
VAT Table	0.63	0.80	0.63	0.98	0.88
Other	0.63	0.80	0.63	0.98	0.88
Average	**0.62**	**0.80**	**0.65**	**0.99**	**0.83**

The model demonstrates strong performance in most header group classes but struggles with supplier name and address, achieving F1-scores of 0.22 and 0.16. Overall, the model performs well with an average F1-score of 0.826, except for the invoice summary VAT class, which has a precision of 0.58. The model is moderately successful in classifying table groups with an F1-score of 0.73 and achieves a 0.65 F1-score for predicting VAT rates on invoice lines. While it performs well for most VAT table classes, it struggles with the VAT subtotal and VAT class, with F1-scores of 0.52 and 0.02, respectively. Non-grouped classes generally have high metrics, suggesting that further training could improve results. When evaluated using the LayoutLMv1 model for practical applications, the model significantly reduced the average invoice processing time by 46.1% in a test involving 20 invoices and 3 participants, despite the small dataset size.

C. LayoutLMv3

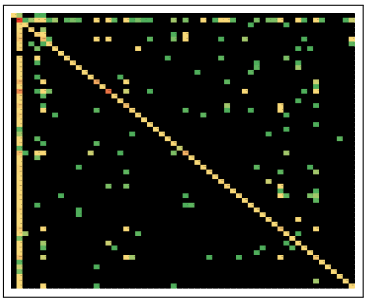


Fig. 4. Confusion matrix of LayoutLMv3 model on 58 classes

After the fine-tuning process, the LayoutLMv3-based model underwent the same tests as the other models, leading to the results shown in Figure 4 and Table IV. When examining the confusion matrices, it can be seen that the predictions of LayoutLMv3 are mainly concentrated on the main diagonal, similar to LayoutLMv1. However, it is observed that the model frequently makes incorrect predictions, especially in the false positive direction and secondarily in the false negative direction, when evaluating the class "Invoice Payment Info". As can be seen from Table IV, this situation significantly reduces the overall recall value of the model and decreases the model's average success. When the classes, excluding the invoice payment information class, are evaluated, it is found that the model has very few erroneous predictions. When these results are evaluated, the model outperforms other models on this task. To obtain more consistent results, the model's training dataset can be enriched with invoices containing examples from the invoice payment information class. No significant inference can be made when examining the model's success by looking at the average success of class groups in Table IV. It can be said that the success of the model is evenly distributed across all invoice regions. While the precision value is high, the reason for the lower recall and F1-score values compared to the precision value is the high number of false negative predictions.

TABLE IV
LAYOUTLMV3 GROUPED METRIC AVERAGES

Group	Precision	Recall	F1	Accur.	Balanced Accur.
Header	0.98	0.68	0.79	0.99	0.84
Table	0.94	0.55	0.68	0.99	0.70
Summary	0.96	0.56	0.70	1.00	0.76
VAT Table	0.97	0.62	0.73	0.99	0.75
Other	0.97	0.62	0.73	0.99	0.75
Average	**0.95**	**0.61**	**0.72**	**0.99**	**0.80**

The LayoutLMv3 model demonstrates remarkable performance in the header group, achieving an F1-score of 0.79. However, it encounters challenges with the supplier phone number class, which has an F1-score of 0.64. Precision values are exceptionally high, averaging 0.98, while recall averages 0.68. However, some classes, such as invoice due date and supplier phone number, are performing below average. In the summary group, most classes perform moderately. However, the invoice summary total class stands out with a recall of 0.38 and an F1-score of 0.53, suggesting frequent misclassification. For table group classes, precision is high but recall is low, indicating accurate but incomplete predictions. The VAT table group achieves high precision but moderate recall, indicating frequent incorrect predictions. Non-grouped classes generally exhibit above-average success, although retention rates often decrease to moderate levels. Overall, although precision metrics are excellent, recall values are notably lower, resulting in lower overall recall metrics for the LayoutLMv3 model.

D. Comparison of Models

After the training of all models was completed, the obtained metrics were used to compare the models. In Figure 5, the average precision, recall, F1-score, and accuracy values are displayed for all three models. When comparing based on precision value, as seen in Figure 5, LayoutLMv3 demonstrates higher accuracy with a value of 0.948. When considering the recall value, the LayoutLMv3 model falls behind the LayoutLMv1 model with a recall value of 0.611 compared to 0.800. When evaluating based on the F1-score, which is the harmonic mean of precision and recall values, LayoutLMv3, the most successful model for precision, is also the most successful model in this comparison. The GCN model is the least successful model in terms of F1-score, while the LayoutLMv1 model is ranked second with a slightly smaller difference compared to the LayoutLMv3 model, primarily

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because of its low precision value. When considering the accuracy value, the results of models show a similar ranking of LayoutLMv1, LayoutLMv3, and GCN. However, as mentioned earlier, having a large number of classes increases the true negative value, which reduces the range of possible accuracy values. This makes it challenging to compare and evaluate the success of models based solely on this value. Taking this into account, utilizing the F1-score, which is influenced by precision and recall values, will yield a more precise outcome for comparing the models.

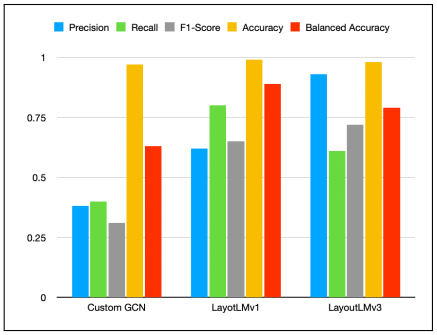


Fig. 5. Comparison of precision, recall, f1-sore and accuracy over all three models

VII. Conclusion

In conclusion, we developed an organizational method for labeling invoices, enhancing data extraction processes and setting a foundation for future studies. Second, we expanded the classification of invoice data into 57 distinct classes, enabling a detailed analysis and understanding of invoice data, which surpasses the typical focus found in existing studies. Finally, we tested various deep learning architectures, including LayoutLMv1, LayoutLMv3, and a custom-made Graph Convolutional Network (GCN). Despite the lack of a pretraining phase, compared to the previous two models, the GCN model demonstrated promising outcomes, suggesting a potential in this domain. Our findings show that transformer-based LayoutLM models, particularly LayoutLMv3, were mostly successful, allowing to contribute to the automatizing of invoices digitization.

VIII. Acknowledgment

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